

Robot Hands Are Learning to Feel!

From: IEEE Spectrum Mined: 2026-05-04 3 min read

Have you ever tried to pick up a delicate cookie without breaking it? Your fingers feel how hard to squeeze. Robots struggle with that because most don't have a sense of touch. But that's changing! A company called DAIMON Robotics has created a special **sensor** that fits on a robot's **fingertip**. It's packed with over 110,000 tiny sensing points like having super-sensitive skin. This lets the robot feel **textures**, **slipperiness**, and how hard it's gripping something. To help robots learn faster, DAIMON built a giant collection of data, called Daimon-Infinity. It's the world's largest **dataset** for robot touch, with millions of hours of information from over 80 real-world tasks like folding clothes or working on a factory line. They even shared 10,000 hours of it for free so other scientists can use it. Prof. Michael Yu Wang, the company's co-founder, says that giving robots a sense of touch is crucial. "Data scarcity remains a primary **bottleneck** in robot learning," he explains. By combining touch with vision and language, robots can understand the world more like we do. In the future, touch-enabled robots might work in hotels, convenience stores, or help at home. So next time you see a robot, imagine it one day being able to feel a high-five!

Word Treasure

sensor -- A small device that can detect things like pressure, temperature, or touch.

fingertip -- The very end of a finger, where the sense of touch is most sensitive.

textures -- The way a surface feels, such as rough, smooth, or bumpy.

slipperiness -- How easily something slides in your grip, like a wet bar of soap.

dataset -- A large collection of information, often used to teach computers or robots new skills.

bottleneck -- A narrow point that slows everything down; here, not having enough touch data slows robot learning.

Background you need

DAIMON Robotics (a company that builds advanced robot hands) has created a sensor that acts like super-sensitive skin for robots.

For a long time, robots could see and hear but not feel--so picking up a delicate cookie without breaking it was almost impossible.

A huge collection of touch data called Daimon-Infinity now teaches robots how to handle things gently, using millions of hours of practice from tasks like folding clothes.

Break it down

WHO: DAIMON Robotics and co-founder Prof. Michael Yu Wang.

WHAT: They built a fingertip sensor with over 110,000 touch points and the world's largest touch dataset.

WHERE: At DAIMON Robotics' labs (exact location not given in the article).

WHY: To give robots a sense of touch so they can learn faster and do delicate jobs in homes, stores, and factories.

Why it matters

When robots can feel, they might one day help us with chores, assist older people, or work in hospitals--making everyday life easier and safer.

Test what you remember

1. Which company created the special fingertip sensor?
 - ☐ (A) NASA
 - ☐ (B) DAIMON Robotics
 - ☐ (C) Apple
 - ☐ (D) Sony
2. How many sensing points are packed onto the fingertip sensor?
 - ☐ (A) About 1,000
 - ☐ (B) Over 110,000
 - ☐ (C) 10,000
 - ☐ (D) 1 million
3. What is the name of the world's largest dataset for robot touch?
 - ☐ (A) Touch-Base
 - ☐ (B) Daimon-Infinity
 - ☐ (C) Sense-Data
 - ☐ (D) Robo-Touch
4. According to Prof. Michael Yu Wang, what is a primary bottleneck in robot learning?
 - ☐ (A) Lack of batteries
 - ☐ (B) Data scarcity
 - ☐ (C) Too many sensors
 - ☐ (D) Slow computers
5. How many real-world tasks did the dataset collect information from?
 - ☐ (A) 10
 - ☐ (B) Over 80
 - ☐ (C) 100
 - ☐ (D) 50
6. How many hours of data did DAIMON Robotics share for free?
 - ☐ (A) 1,000
 - ☐ (B) 10,000
 - ☐ (C) 100,000
 - ☐ (D) 5,000

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: B 6: B

Questions to think about

1. **Robot developer:** Engineers see touch as the missing piece that will finally let robots handle soft, fragile objects and work right next to people.
2. **Skeptic:** Some people worry that even touch-smart robots could be too expensive or might still make mistakes, delaying their use in real homes.

MY THOUGHTS (at least 20 words):